

SUB-MILLISECOND EVENT-VISION SOPC PIPELINE FOR REACTIVE UAV CONTROL

A Master's Thesis Proposal in Computer Engineering

Focus area: reconfigurable computing and embedded systems · Application domain: autonomous aerial robotics

ABSTRACT

Frame-based vision pipelines running on GPU/NPU platforms deliver perception in tens of milliseconds with significant variance, a bottleneck for vision-guided drones that must sense and react in tight timing budgets. This thesis proposes a system-on-chip (SoPC) pipeline that processes event-camera streams in streaming hardware, achieving sub-millisecond, deterministic sensor-to-decision latency. The work establishes, both analytically and experimentally, how this latency reduction and determinism raise the achievable closed-loop bandwidth of reactive UAV control. The contribution is a co-designed streaming architecture on a Zynq UltraScale+ platform, evaluated against GPU and NPU baselines on latency, power, and integration effort, and validated through recorded data, hardware-in-the-loop simulation, and a controlled indoor flight.

1. MOTIVATION AND PROBLEM

Vision-guided drones face two coupled constraints. First, frame-based cameras impose exposure windows and rolling-shutter artifacts, corrupting exactly the information a reactive controller needs in high-contrast or fast-motion scenes. Second, the accelerator stacks used for perception — GPUs and NPUs — insert buffering, batching, and driver-queue latency that is large and variable, typically 10–100 ms. In feedback control, this delay and jitter shrink the phase margin, capping the gain and disturbance rejection the loop can sustain. This makes perception latency a first-order control-engineering problem, not merely a throughput concern.

Event cameras offer a different sensing model: asynchronous per-pixel brightness-change events with microsecond timestamps, high dynamic range, and no motion blur. Their natural output is a stream, which maps directly onto streaming hardware. An FPGA can process that stream with bounded, composable latency that is largely load-independent, whereas GPU/NPU latency scales with batch and queue occupancy. This combination — an event sensor feeding a streaming SoPC pipeline — is the technological core of the thesis.

2. RESEARCH QUESTIONS

1. What end-to-end latency distribution and worst-case bound can a streaming SoPC event-vision pipeline achieve from camera ingress to advisory output under sustained load?
2. How does perception latency and its variance affect closed-loop stability and disturbance rejection, and what response improvements are unlocked at sub-millisecond latency?
3. How does the proposed pipeline compare against GPU and NPU baselines on matched accuracy and input rate, in latency, power, and integration effort?

3. NOVELTY

Prior work treats FPGA vision accelerators as throughput engines or compares fixed-CNN inference across hardware. This thesis takes a different position: it treats the latency-determinism of the perception pipeline as the independent variable and the closed-loop control outcome as the dependent variable. The novelty lies in (i) a complete streaming event-to-advisory pipeline co-designed in HLS and RTL on a production SoPC, (ii) a quantitative latency-to-stability result linking measured latency distributions to achievable control bandwidth, and (iii) an honest, application-level comparison with GPU/NPU baselines that reports where the FPGA loses as well as where determinism wins. The claimed contribution is not "FPGA is faster," but a systems result: deterministic perception enables control behaviour that frame-based pipelines cannot reach.

4. TECHNICAL APPROACH

The pipeline is implemented on a Zynq UltraScale+ / Kria-class system-on-module with a Prophesee GENX320 event camera attached to the programmable-logic fabric. The streaming data path — event pre-processing, time-surface accumulation, optical-flow estimation, and occupancy/obstacle-map update — runs entirely in hardware on the camera clock; only a low-rate aggregated advisory crosses into software. The critical stages are implemented as Vitis HLS kernels with RTL streaming interfaces, giving a mixed HLS+RTL design that keeps the timing-critical path in hardware while preserving development velocity. A control-barrier-function (CBF) safety filter bounds a minimum braking distance from the measured worst-case latency. All stages carry cycle-accurate timestamps to support worst-case-latency accounting.

STAGE	IMPLEMENTATION	LATENCY BUDGET (TARGET)
Event pre-processing (noise, hot-pixel, timestamp sync)	RTL streaming	< 50 μ s
Event accumulation / time-surface generation	HLS kernel	< 100 μ s
Optical-flow estimation	HLS kernel, streaming	< 300 μ s
Occupancy / obstacle-map update	HLS kernel	< 200 μ s
CBF safety filter + advisory encode	ARM + RTL assist	< 50 μ s

5. CONTROL ENGINEERING DIMENSION

Control engineering enters the thesis as a first-class partner discipline, not an afterthought. A time delay in the feedback loop adds phase lag proportional to the crossover frequency, so reducing perception latency from tens of milliseconds to sub-millisecond values raises the bandwidth a loop can sustain before its stability margin is violated. The thesis quantifies this relationship: it measures the latency distribution, translates it into achievable bandwidth and disturbance rejection, and demonstrates the benefit experimentally with a reactive-braking loop guarded by a CBF safety filter. The scope is deliberately bounded to a tractable but rigorous formulation — a latency-to-stability systems argument with experimental validation — avoiding the depth of a full control-theory programme.

6. EVALUATION AND VALIDATION

- **Bench:** end-to-end latency distribution (mean, p95/p99, observed max) and power on FPGA, GPU (Jetson-class), and NPU (Hailo-class) at matched accuracy and input rate.
- **Hardware-in-the-loop:** replay recorded event streams with PX4/SITL; sweep an artificial latency offset to demonstrate the stability/response relationship; measure avoidance reaction time and CBF margin preservation.
- **Flight:** one controlled indoor VLOS demonstration of reactive avoidance; field validation only after bench and HIL gates pass.

7. SCOPE BOUNDARIES

The system emits advisory outputs only and never directly commands flight controls, keeping the safety and regulatory burden out of the thesis. A secondary, clearly-scoped extension demonstrates event-based precision landing to show the latency advantage transfers to a second control loop. A custom PCB is explicitly deferred; development uses a production-oriented system-on-module.

8. EXPECTED OUTCOMES

A streaming SoPC event-vision pipeline with measured, bounded sub-millisecond latency; a publishable latency-to-stability result for reactive UAV control; and a fair, reproducible comparison with GPU and NPU baselines. The work targets an FPGA/embedded-systems venue, with an extended UAV-journal version after field validation.